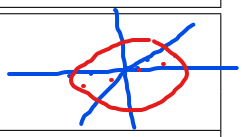
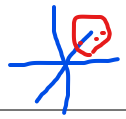
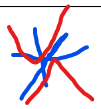
Worksheet 2 - 12.5, 12.6

Complete in Discussion

1. Fill out the table below for each of the common surfaces.

Name	extra description	Equation	Sketch
Sphere	centered at origin with radius r	x^2+y^2+z^2=r^2	
Sphere	centered at (a,b,c) with radius r	(x-a)^2+(y-b)^2+ (z-c)^2=r^2	
Ellipsoid	centered at origin	ax^2+by^2+cz^2=r^2	
Circular cylinder	centered and opening along z-axis with radius r	x^2+y^2=r^2	
Paraboloid	centered and opening along z-axis with vertex at $(0,0)$	z=x^2	
Elliptic Paraboloid	centered and opening along z-axis with vertex at $(0,0)$	z=x^2+y^2	
Cone (True Cone)	centered and opening along z-axis	z=+-sqrt(x^2+y^2)	
Hyperbolic Cylinder		x^-y^2=c	
Parabolic Cylinder		z=x^2	

x=3t-2+1=3t-1
y=2t+1+3=2t+4
z=t+4+2=t+6
<3t-1,2t+4,t+6>

2. Find the equation of the line that goes through the point $(1,3,2)$ and is parallel to the line $\langle 3t-2, 2t+1, t+4\rangle$. Given both the vector equation and the parametric equation.

3. Find the equation of the plane containing the points $A=(1,2,3), B=(2,3,4)$ and $C=(3,4,6)$.

AB = <1,1,1>
AC = <2,2,3>
AB x AC = n = <3-2,-(3-2),2-2>
n = <1,-1,0>

n * AC = 1(x-1)-1(y-2)+0(z-3) = 0
0=x-1-y+2
0=x-y+1
-1=x-y

Extra Quiz/Exam Practice

4. What region in $\mathbb{R}^3$ do each of the following equations represent? Explain your reasoning.

   (a) $x^2 + y^2 = 49$

   (b) $x^2 + y^2 = 49, z = 1$

   (c) $x^2 + z^2 \leq 49$

   (d) $x^2 + y^2 + z^2 = 49$

   (e) $0 \leq x^2 + y^2 + z^2 \leq 49$

5. Find the equation of a plane that passes through the origin and contains the line $x = 2t, y = 3t, z = 3t - 1$. Is there only one plane that satisfies these requirements?

6. Give an example of at least four different families of surfaces where all of traces parallel to the xy-plane are circles. For each "family" (e.g. spheres) give an example of a specific member of that family (e.g $x^2 + y^2 + (z-3)^2 = 36$)). Explain how you know each trace is a circle. Then explain how you can tell the families apart from each other.

Challenge Questions

7. Find the intersection of the planes $x + 2y - 3z = 1$ and $x - y + z = 1$ two different ways. (For example, for one, you can explain why the direction vector of the line of intersection must be orthogonal to both the normal vectors of these planes and find it using the cross product. Then you just need to find a point on the line (so on both planes). For a second, you could set both equations equal to x, for example, and then equal to other, then solve for y and z in terms of a single parameter.)

8. Where does the line through $(1, 2, 1)$ and $(2, 1, 4)$ intersect the plane $2x + 3y + z = 10$?

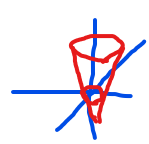
9. For the surface below, reduce the equation to standard form, classify the surface and sketch it.

$$x^2 + y^2 - 2x - 6y - z = 10$$

7)

8)
A = (1,2,1)
B = (2,1,4)
r(t) = (1+1t,2-1t,1+3t)
2(1+t)+3(2-t)+(1+3t)=10
2+2t+6-3t+1+3t=10
2t+9=10
2t=1
t=1/2
r(1/2)=(1.5,1.5,2.5)
= intersection point

9)
x^2-2x+1+y^2-6y+9-z
=10+1+9
(x-1)^2+(y-3)^2-z=20
the function
represents a cone
centered at (1,3,0)



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